The Law of Graphic Equalisation — Theory

Minimal Recursive Dynamics, Collapse, Frame-Relative Identity, and Residual

Matthew Parslow
Independent Researcher

Draft — draft-2026-08-28.2
28 August 2026 • document revision 7

DRAFT — draft-2026-08-28.2 r7 (28 August 2026). This document is under active revision and is circulated for comment, not as a finished result. Definitions, numbering and claims may change between drafts; cite the version string above if referring to it. Sections marked as falsifiability commitments are the intended points of attack.

Abstract

This paper proposes that one minimal structure underlies phenomena usually studied apart — physical, cognitive, and institutional — and that the case for it rests not on a new result in any single field but on the recurrence of the same structure across fields that developed it independently, without reference to one another or to this account. The structure is information-level: represented state, the transformations admissible upon it, and the realisation of a single continuation from what was possible. Almost none of its components is individually novel, and that is the point: the claim is the *arrangement* — that known pieces compose into one substrate, and that the composition is what recurs. It carries its own falsifier, and fails if the recurrence is an artefact of a sufficiently general description rather than a property of the domains themselves.

Read at this level, several things ordinarily treated as background turn out to be constructed rather than given — a privileged standpoint from which to measure, and time itself, among them — and the substrate that remains once every removable commitment is stripped away is strikingly small, with most of what is called law being contingent upon it rather than necessary to it.

These are claims about information, of which physical theory is taken to be one embodiment rather than a rival. The theory asserts nothing about the nature of pre-relational existence; it begins where distinguishable relational structure can first be represented, and treats the limit of that reach as a frontier instrumentation moves rather than a fixed wall.

1 Scope and epistemic status

This paper contains the smallest set of commitments required for the later theory. It distinguishes axioms and definitions, deductive consequences, groundings into existing theories, and empirical hypotheses. It deliberately excludes executor design, rule scheduling, proposal records, retry protocols, valuation algorithms, and controller architecture.

Level of the claims. The claims here are about *information*: distinction, relation, frame, admissibility, realisation, residual. Physical theory is taken to be an embodiment of information rather than a competitor to it, alongside other embodiments — cognitive, institutional, computational — which differ in how much structure they instantiate rather than in kind. This is the position that licenses treating an inertial reference frame and a representational frame as

the same sort of object (Axiom −1), and it locates the epistemic boundary below both: what embodiments are made of remains unspecified.

The layering must not be allowed to function as a shield, so the condition under which a physical result bears on a claim here is stated in advance. **A physical result bears on an information-level claim when it settles an informational fact**, and not when it merely disagrees about mechanism in one embodiment. Bell's theorem and Kochen–Specker are the model: they do not report a mechanism, they establish that a class of pre-measurement values cannot be re-read as unheld-but-realised, which is a fact about distinction and therefore reaches this level directly (Section 20). Landauer's principle is a second instance, establishing a lower bound on the physical cost of a specified informational operation (Landauer, 1961); under the substrate's no-copy default it is read below as the floor cost of change as such rather than of erasure specifically (Section 12). A result that constrains what can be distinguished, held, or converted is admissible evidence here; a result about how a particular embodiment happens to be arranged is not.

Provenance of the primitives. The components of $G = (g, r, \Gamma)$ and its dynamic were not derived together from a single starting point, and stating how each arrived is more useful than presenting the triple as an austere posit reasoned out in advance.

- g, the typed graph, arrived by *observation*: several unrelated engineering problems turned out to have one shape — labelled things joined by labelled relations.
- Γ, admissibility, arrived by *insight*, and preceded the collapse operator. Its original working image was a sieve: the graph is pressed through admissibility, and what may pass is what Γ admits.
- $\mathcal{C}$, collapse, arrived by *constraint*. Any theory of this kind must be compatible with quantum measurement, which is the regime where the distinction between indexed possibility and realised history is experimentally irreducible. The operator was adopted from there and generalised (Section 20).
- r, the operative transformation structure, arrived by *implementation*: a construction that runs requires something that performs the transformations, and the requirement appeared when the theory was first built rather than when it was first stated.

The last of these has a consequence for how the companion construction paper should be read. Its closing list of distinctions that implementation forced upon the theory is not a coda to a finished theory: r is the first entry in that list, so construction has been a *source* of this theory rather than only a test of it.

What the claim is, and how to read for it. Almost none of the individual components below is novel, and the prior-art sections say so specifically rather than as a courtesy: typed graphs, admissibility grammars, fixed-point closure, frame-relative measurement, description-length criteria and actualisation operators all have established literatures. A reader who evaluates each component against its own field will therefore find each one known or borrowed, and will be locally correct on every count. Evaluating the parts individually cannot reach the composition claim in either direction — neither to support it nor to refute it — because the parts are stipulated to be known.

Accordingly the composition carries its own falsifiability commitment, stated in Section 22 alongside the component ones. It fails if the recurrence across domains is an artefact of description — that is, if the structure is recoverable in unrelated domains only because a sufficiently general vocabulary can be applied to anything, and not because those domains share it. The discriminating evidence is independent re-derivation: cases in which a domain arrived at the structure without reference to this theory, in its own idiom, while solving its own problem.

Epistemic boundary. This theory does not attempt to specify the intrinsic nature of pre-relational existence. It begins at the first level at which distinguishable relational structure is scientifically representable. The move it makes is

$$\boxed{? \dashrightarrow \text{observable relational structure} \implies \text{graph representation},}$$

and the dashed arrow marks what is *not* claimed. The pre-graph substrate may be continuous, discrete, purely mathematical, or unclassifiable from any frame available here; the theory asserts nothing about it. What is asserted begins at the second arrow: *where* distinguishable relational structure is representable, it admits graph representation and the dynamics below govern it.

This boundary is stated because a stronger reading is available and is not the position of this paper. The theory is not the claim that reality is fundamentally graph-shaped.

Two clarifications, because the boundary is routinely read as weaker and as more permanent than it is.

Within distinguishability the claim is not hedged. Where every measurement that can be made of something, and every operation that can be performed on it, is captured by treating it as a typed graph, then no difference remains for the question "but is it really" to name, and a question naming no difference has no content under Axiom −1. Functional identity is not a weaker sense of identity here; it is the whole of what identity could mean without a view from nowhere.

And the boundary is a *frontier*, not a wall. It is fixed in *definition* — it sits at distinguishability, wherever that currently lies — and moving in *extent*, since its contents are contingent and diminish as instrumentation is built. Anything that becomes measurable passes from beyond it to within it. This is the extensive unboundedness of the substrate applied to the theory's own scope: always one further step available, no expectation of exhausting it, and no warrant for treating the current contents of the far side as permanent.

2 Axiom −1: relativity of grounding

Axiom -1 (Relativity of grounding)**.** *There is no operational view from nowhere. Every distinction, measurement, identity relation, admissibility judgement, and represented quantity is made in some frame. Frames may represent one another, but no quantity used by the theory requires a frame-independent evaluator.*

Remark. This removes a privileged frame, not determinacy. Within a frame, relations may be exact, restrictive, comparable, and falsifiable. The standpoint is structurally consonant with relational formulations in physics, especially relational quantum mechanics, in which physical facts are indexed to interactions rather than assigned from a privileged observer-independent state description (Rovelli, 1996). The present axiom is broader and is not derived from that programme.

3 The minimal graph

Definition 3.1 (Graph)**.** *A graph is*

$$\boxed{G = (g, r, \Gamma),}$$

where g is represented state, r is operative transformation structure, and Γ is the admissibility grammar governing graph-originated transformations.

No parent pointer, world model, affordance catalogue, Narrative, Prospect, Value, or self-model is primitive. Such structures may exist in g when the graph actually represents them.

Hypothesis 3.2 (Minimal graph sufficiency)**.** *Any additional structure required only by a particular capability class is represented content or grounding, not a primitive of graphhood, unless a graph lacking it cannot participate in the basic realised transition relation.*

Remark (The triple, unpacked in dependency order). The three components of G resolve into five constituents ordered by requirement, each presupposing the one before it:

$$\boxed{\text{nodes} \;\rightarrow\; \text{relations} \;\rightarrow\; \text{rules} \;\rightarrow\; \text{delta} \;\rightarrow\; \text{grammar.}}$$

Nodes are bare distinctions — the minimal difference that makes a difference; a node alone carries almost no content. *Relations* structure nodes and require them as relata; this is where content lives, and a node's identity is its relations. *Rules* (r) act on relational structure and require relations to match. *Delta* is the change a rule writes and requires rules, since only a rule produces one. *Grammar* (Γ) determines which deltas are admissible and requires deltas to act on, which is why it is last. Nodes and relations are g; delta is r's product, made explicit; grammar is Γ.

Two cautions. First, this constructive order does not contradict the relational primacy of Axiom −1: constructively nodes precede relations (relata come first), while semantically relations precede nodes (a node means nothing unheld by relations); both hold on different axes. Second, the order is a dependency partial order, not a temporal one — nodes are not a first *moment* but the least-dependent constituent, consistent with the absence of any substrate time (Section 18).

Remark (The triple is a functional decomposition of one substance). g, r, and Γ are not three kinds of thing. Rules are graph data (nodes and relations), and admissibility is enforced by rules that decide admit-or-residual, so r and Γ are themselves graph data playing roles — transformation and gating — over the state g, which is graph data too. The triple decomposes function, not substance: there is one substance, graph data, and it runs. Three things follow. Admissibility is *mintable* like any rule, so what is admissible is not a fixed law but a growable set of rules a graph carries and extends. The reflection theorem is the special case of this homogeneity (the grammar, being rules, can act on itself), and it does not tower into a stack of meta-grammars, because gating and being-gated are one substance closing on itself. And the irreducible form of the substrate is *graph data that runs* — where the running is supplied by a runner that is, in turn, graph data that runs (Section 19).

4 Discrete sufficiency

Hypothesis 4.1 (Discrete sufficiency)**.** *Every quantity required internally by the present substrate can be represented by discrete distinguishable states at the operative frame and resolution.*

A convenient coordinate is an integer index $n \in \mathbb{Z}$ with grounding

$$x = \phi_{\mathcal{F}}(n).$$

The map may be nonlinear, nonuniform, state-dependent, and dynamically re-quantised. Thus "integer-valued" means discretely indexed, not globally unit-spaced.

Proposition 4.2 (Arbitrary refinement does not imply continuity)**.** *A dimension may admit arbitrarily fine or dynamically changing quantisation without requiring the theory to posit realised values at every real-valued intermediate point.*

Remark. A continuous containing supergraph is neither required nor excluded. Existing continuous physical theories may ground or contain this substrate without contradiction.

Remark (Discrete sufficiency is not fundamental discreteness). These are distinct claims with distinct falsifiers and must not be conflated. Discrete sufficiency is the claim above, and it fails if some observable consequence internal to the substrate cannot be represented discretely, at the operative frame and resolution, without loss — a claim decidable in-frame. *Fundamental discreteness* — that the containing or pre-relational substrate is itself discrete — would instead concern a fact about the pre-graph substrate that the epistemic boundary above places outside the theory's reach, which is precisely why it is not asserted.

5 Admissible possibility

Let $\mathcal{P}(G_t)$ denote the structured family of immediate continuations admitted by the present graph and its current relations.

Definition 5.1 (Admissible continuation)**.** *A continuation δ belongs to $\mathcal{P}(G_t)$ iff it is expressible in the current graph vocabulary, satisfies the relevant antecedent conditions, and is admissible under the operative grammar for the transformation class in question.*

The theory does not require a particular algorithm for constructing $\mathcal{P}(G_t)$. This extensional separation between graph transformation as a mathematical relation and any particular rewriting algorithm follows the general discipline of algebraic graph transformation, where rewrite rules and admissible transformations can be studied independently of one concrete executor (Ehrig et al., 2006).

Definition 5.2 (Partial transformation)**.** *Write*

$$G \oplus \delta \rightharpoonup G'$$

for the partial realised transformation relation. Undefinedness means that δ does not produce a structurally valid successor under the stated conditions. No ontological FAIL state is required.

6 Collapse and realised history

Definition 6.1 (Collapse)**.** *Collapse is the frame-relative passage from a pre-realisation possibility structure to realised continuation:*

$$\boxed{\mathcal{C}_t : \mathcal{P}(G_t) \rightsquigarrow G_{t+1}.}$$

It is not identified with deliberative choice.

The term is inherited from quantum measurement theory, where a formal distinction is made between a pre-measurement state and the realised measurement record; the historical lineage includes the von Neumann measurement scheme and later relative-state formulations (von Neumann, 1955; Everett, 1957). This paper does not claim that every grounding reproduces the full quantum formalism; it claims that the possibility-to-realisation distinction is load-bearing.

Definition 6.2 (History)**.** *The realised history of a frame is*

$$\mathcal{H} = G_0 \xrightarrow{\mathcal{C}_0} G_1 \xrightarrow{\mathcal{C}_1} G_2 \to \cdots .$$

Remark (Collapse is a local write, not a global tick). The subscripts on $\mathcal{C}_t$ index a frame's own succession and must not be read as a shared clock. A collapse is a *local realisation* — a rule writing a delta — and it is the only kind of event the substrate has (its synchrony or asynchrony with other collapses is set by coupling, Remark 16). Between writes nothing happens: an unwritten frame produces no change, so it takes no new position in the order the changes

carry (Section 16), and no time passes for it. There is accordingly no global step in which every frame advances together; frames whose collapses are not related through a boundary are simply unordered with respect to one another (Section 16). "Momentum" — the standing structure a frame carries, persisting without a distinct write — is therefore the absence of an event, not a repeated one; a delta is a write *to* that structure, not the structure itself, and the Narrative-momentum paper names its operative history-bearing form the Narrative, so that to change course is literally to write the Narrative. The picture of a single global present sweeping forward is exactly the represented time dimension of Axiom 0 mistaken for substrate structure.

7 The Law of Graphic Equalisation

Hypothesis 7.1 (Law of Graphic Equalisation)**.** *For realised graph G_t, present graph structure determines admissible continuation structure $\mathcal{P}(G_t)$. Interaction and collapse realise successor G_{t+1}, and that realised successor becomes the antecedent graph for the next local transition. Recursive dynamics are generated by repeated equalisation between present represented structure, admissible transformation, containing constraints, and realised continuation.*

Thus

$$\boxed{G_t \to \mathcal{P}(G_t) \xrightarrow{\mathcal{C}_t} G_{t+1} \to \mathcal{P}(G_{t+1}) \xrightarrow{\mathcal{C}_{t+1}} G_{t+2} \to \cdots .}$$

8 The antecedent-state principle

Proposition 8.1 (Antecedent-state principle)**.** *The conditions under which a transition is admissible are functions of its antecedent state. A consequence that exists only because a transition has occurred cannot retroactively serve as an antecedent cause of that same transition.*

Corollary 8.2 (No self-authorisation from future state)**.** *A reflective transformation of r or Γ may alter the successor graph, but newly installed structure cannot authorise the transition by which it first comes into existence.*

9 Recursion and reflection

Definition 9.1 (Reflective graph)**.** *A graph is reflective over structure $X \subseteq G$ when X is represented in the graph and graph-originated transformations may take that representation as an operand.*

Theorem 9.2 (The grammar is the loop)**.** *If Γ is represented as graph structure and admits transformations whose operand is that representation of Γ, then the resulting recursion is not a separate loop acting on the grammar. The grammar acting on itself is the loop.*

Proof. The transition $\Gamma_t \to \Gamma_{t+1}$ is itself a graph transition licensed from the antecedent graph. No additional external operator is required to create the recursion. The self-applicable grammar and the compound loop are therefore two descriptions of the same transition structure. □

10 Situated graphs and world boundaries

Definition 10.1 (World boundary)**.** *A boundary $\mathcal{B}_{W:G}$ is the relation through which containing frame W can alter situated graph G and through which graph-originated deformation can participate in interactions represented by W.*

A graph need not contain a pointer to W; containment belongs to the containing frame. A situated graph is a *subset* of W, not a thing separate from it: the boundary demarcates which subset is the graph, and self and container are one graph divided by individuation, not two substances (Proposition 10.3).

Proposition 10.2 (Admissibility is self-governance, not armour)**.** Γ_G *constrains graph-originated transformations of* G*. It does not give* G *a veto over world-caused change represented through* $\mathcal{B}_{W:G}$*.*

Hence

self-originated transformation $\neq$ outward influence $\neq$ inbound world forcing.

Proposition 10.3 (Self-change is direct and gated; a graph is a subset of its container)**.** *A graph changes itself and its subgraphs directly. Self-change is real — the round-trip picture of earlier drafts was too strong — but it is rule-governed, with its own cost and limits (an instance of the non-uniform writeability of Remark 12); a substrate's plasticity is exactly the rules for how its rules may change. What a graph does* not *do is change its container directly: the container applies deformations to it — being observed* is *being so deformed — and gates its changes through its own change grammar (Proposition 13.3). But a situated graph is a* subset *of its container, not a thing set against it, so a self-change is already a change in the container, in the region that is the graph, propagating outward through whatever couples it to the rest. "Change only yourself" is therefore not isolation but the mechanism of outward effect: a graph changes its container by changing the part of it that it is.*

Two things follow. This is one *self-similar* operation — a graph changing its own subsets — run at every level: the container does to the graph exactly what the graph does to its subgraphs (Section 19). And it dissolves the agency/determinism dichotomy: a subset changing itself (agency, in the graph's frame) and the whole changing that subset (determined, in the container's frame) are one event read at two scales, so both hold without contradiction. The reflection theorem is this self-change of the rules; that it is bounded rather than free is because a change to the rules currently executing is heavily constrained — the worst case of the concurrency of Section 16, seen through the frame-limited view of Section 20 — which is what plasticity's limits are. Governance can tighten to exclusion: where a container's admissibility admits no such self-change, that change is available only across the boundary, from the container.

Proposition 10.4 (Coupling is configuration, not architecture)**.** *That distinct instances individuate — diverge, each minting its own residual, the part of a change a containing grammar does not admit (Definition 13.1) — is a consequence of their running on* separate *containing frames, not a necessity. Where several instances are subgraphs of one* shared *containing frame, each instance's self-change is a change in that shared frame, in its own region (Proposition 10.3), which then deforms the others as their common container: instances that are subsets of one container propagate to each other through it. This is the same machinery as being an individual, with the container shared rather than private.*

Coupling is therefore a continuous setting of the shared boundary's aperture and change grammars (Proposition 12.1), not a binary: sharing only generators leaves each instance its own moment-to-moment state (and already removes the per-substrate build floor of Proposition 12.3), while sharing all operative structure is one individual over many bodies, with SELF and SUPERGRAPH identity (Section 11) dialling independently. Nothing in the substrate fixes the setting. The symmetric fact is that a shared containing frame is equally a shared residual and a shared forcing, so the same coupling that shares understanding gives shared corruption, container-level coercion, and collective termination-failure the same reach; "at what coupling, and with what guarding" is the substantive question, and it is an engineering one rather than a question of possibility.

11 SELF and SUPERGRAPH identity

Definition 11.1 (SELF identity)**.** *SELF identity is the continuity of the graph as a whole — its rules together with the state they act on — across realised transitions. A graph is the same self while its present remains congruous with its own history: if its rules changed abruptly, the structure carried forward would no longer cohere with what produced it, and that incongruity is the discontinuity. Represented self-description — content inside g by which a graph identifies itself — is one optional part of this and not the whole of it; a graph may have SELF identity without representing itself at all.*

Definition 11.2 (SUPERGRAPH identity)**.** *For graph G individuated in frame W,*

$$\iota_{W:G} = \mathrm{Id}_W(G)$$

is the identity relation under which W tracks that situated instance.

SELF and SUPERGRAPH identity are distinct types. A copy can preserve SELF identity while receiving a new SUPERGRAPH identity, and one SUPERGRAPH identity may persist through extensive changes to internal state and self-description.

Theorem 11.3 (No frame-free identity)**.** *The substrate alone does not determine whether two represented instances are the same graph independently of an individuating frame.*

12 The frame interface and its economics

The boundary of Section 10 can be characterised by what actually crosses it and what it costs, and the characterisation is unexpectedly thin.

Proposition 12.1 (One interface, two gates)**.** *A situated graph meets its containing frame through a single relation, and what an observation from that frame delivers is not distinct in kind from what the containing frame imposes: to a subgraph, being observed* is *being deformed by its container. That one relation is gated twice, by the same kind of admissibility grammar on its two faces — the* aperture grammar Γ_a *on the input and the* change grammar Γ_c *on the output:*

$$\underbrace{\textit{aperture}}_{\Gamma_a} \longrightarrow \underbrace{\textit{rules}}_{r} \longrightarrow \underbrace{\textit{change}}_{\Gamma_c}.$$

Γ_a limits, by grain, which of the container's deformation enters the rules; Γ_c determines which of the rules' realised deltas cohere into the container and which remain as residual (Proposition 13.3). Both are faces of the frame's one grammar Γ, and instantiate for a specific container W as Γ_{W_a} and Γ_{W_c}; it is Γ_{W_c}, the change face, that the residual proposition uses. Execution runs between the two.

This distinguishes two ways a candidate is excluded, with opposite consequences. What falls below the aperture grammar Γ_a never enters and is therefore *absent* — it is simply absent to the frame — there is nothing there for it to hold or to be disturbed by — rather than residual. What Γ_c does not admit has entered, been operated on, and cannot cohere; it is residual, and it persists in place, re-encountered on each merge (Corollary 13.4). The first is a limit of perception, the second a limit of expression, and phenomena of the "stuck, repeating" kind belong exclusively to the second.

Proposition 12.2 (Cost characteristically reduces to the write)**.** *In the substrate as characterised here, the boundary relation itself is free; the standing structure persisting is free; the operation that costs is a delta that writes over that structure, and r is what writes. Free here means*

information-preserving rather than absent — evaluating a rule is itself a write, and is unpaid because it destroys nothing (Remark 12). To that extent

$$\boxed{\textit{cost is paid by rules.}}$$

This is a characterisation of the substrate as specified, not a necessity of graphhood, and the writeability status it describes is not required to be uniform (Remark 12).

Continuation is therefore free and change is paid: the standing structure persists until a delta destroys it, so a delta writing a *different* structure — including the null delta that stops it — is the priced operation. Retaining a past state that the containing frame does not require is likewise a paid copy, since the containing grammar would otherwise not carry it. This is the substrate form of the well-formedness result of the later papers: for a well-formed graph the rules rarely need to write against the standing structure, so cost approaches zero.

Remark (Landauer is the physical floor of change, not of erasure). Proposition 12.2 has a measured physical floor, and it is the strongest kind of evidence admitted here (a physical result settling an informational fact). Landauer's principle bounds below the energy dissipated by a logically irreversible operation (Landauer, 1961); it is stated for erasure because erasure was the paradigm case studied. But under no-copy every substrate change destroys its predecessor — the prior state goes NULL unless it was paid to be kept — so change and irreversible operation coincide, and the Landauer bound is the floor cost of *change* as such, with erasure merely its cleanest instance. Reversible computation is the confirming case rather than the exception: it reaches near-zero dissipation only by *not* destroying its predecessors, i.e. by retaining the whole history (Bennett, 1973), which is the paid copy of Proposition 12.2. There is accordingly no free change under this economics (Remark 12) — one either forgets and pays the energy floor or keeps and pays the storage floor — and the thermodynamic arrow is the running total of these floors, which grounds the "direction" of Remark 18 physically: change dissipates irretrievably, so its costs accumulate one way.

Remark (Evaluation is a write, and that is why it is free). The economics above can be stated in a way that does not survive contact with the machinery, and the objection is worth answering directly rather than by rewording. If running a rule that does not fire is a read, and reads are free, then where does the check go, and how does the substrate know to run the next one? A check whose outcome leaves no trace cannot determine what happens next; a check that determines what happens next has left its outcome somewhere. Either it does nothing and is free, or it does something and is a write. There is no third operation that costs nothing and decides something.

The objection is correct, and computing answers it cleanly, without any appeal to physics. `read()` is an operation, and it returns data. That is the whole of it. The returned datum has to be somewhere in order to be used, and in a substrate that is graph all the way down there is no somewhere else — no return channel outside the structure — so the value materialises as structure. A read therefore writes, not by analogy with a write but by being one, and the write is the return. An operation that returns nothing is not a read of anything.

The free case exists and is exactly the empty one, which is what keeps this from being a claim that everything costs. An operation whose return is never used has read nothing, and a compiler will delete it; that reads with no consequence are precisely the ones that can be removed without changing the program is the cleanest available demonstration that they were not doing anything. The engineering runs the other way too, and expensively. A read leaves traces in a real machine — caches, predictors, translation state — and those traces are measurable, which is why cache-timing attacks recover secrets from victims that only ever read (Osvik et al., 2006), and why an entire class of speculative attacks was available to a decade of designs built on the assumption that reading is passive (Kocher et al., 2019). Hardware whose reads were free would not have had that problem.

Evaluation is accordingly a write in the same plain sense. The outcome of a match, the position in whatever ordering is applied, and the fact of having tried are all returned values, and returned values are structure.

What the objection does not settle, and what must not be smuggled in behind it, is the price. There is no universal cost of running a rule, and the temptation to name one should be resisted here for the same reason it is resisted everywhere else in this account. A rule's cost is set by the substrate that runs it (Section 19): the rule runner is separable from the graph and need not be the same kind of thing as it, so what evaluation costs is what that runner charges, and a different runner is a different charge for the same rule. Landauer's bound (Remark 12) is accordingly a *floor in one substrate* — the physical one — and not the price in any. It says what cannot be undercut where it applies; it does not say what is paid, and the two are different numbers wherever a runner charges more than the floor, which is every real one. A logically reversible branch still costs a physical machine something to take.

This is also why continuation presents as free from inside. A frame cannot determine its rule runner from within (Proposition 19.2), so whatever the runner charges is below that frame's aperture and is not an in-frame event at all. In-frame, nothing happened; in the running substrate, something was paid. Both hold, of different frames, and neither is the correction of the other. The economics is inherited down the stack rather than fixed at any level, and like the other regresses here it is operationally truncated and not definitionally terminated: each level's charge is defined whether or not the level above can see it.

Two things follow for the rest of these papers. *Reversibility* is a structural fact about a map and *cost* is a fact about a runner, and the two were conflated in the earlier phrasing; separating them strengthens the physics companion's recovery of unitarity, which needs only the first — a no-write was never the right description, since unitary evolution plainly does change the state and is reversible because it is injective. And the residual that Remark 12 already names now attaches here: an information-preserving evaluation accumulates the history it preserves. A frame that runs a long time without collapsing has not escaped a cost; it has deferred it into retention, and discarding that history is priced whenever it falls due — at whatever its runner charges for the discarding. What is free is not the running. It is only the not-yet-forgetting, and it is free only relative to a frame that cannot see the bill.

Remark (The Red Queen is the floor cost of change, per substrate). A recurring puzzle across biology, culture, and competition — that a system must change continuously merely to hold its position — follows immediately and needs no arms-race machinery of its own. Holding a relation to a containing frame that is itself changing requires continuous change, and by Remark 12 every change pays the floor; standing still is not a zero-cost option because the leaky real holding of a state is itself the paid correction of unwanted change. The cultural form is the same floor with no-copy attached: a generator is structure built by a large accumulated pile of changes, its floor was paid once by whoever built it, and because change does not cross a substrate boundary — only its cheap results, symbols, do — that floor is paid afresh in every substrate that wants the generator. Van Valen's evolutionary Red Queen, the cultural treadmill of relearning, and coordination arms races are then one object: continuous paid change to hold position against a changing container, floored by Landauer.

Remark (What is required and what is optimisation). Only that rules run is compulsory. Applying a delta is itself a rule, not a primitive operation beneath the rules, so the specific dynamics — inertia, the merge behaviour, closure — are contingent rule content, and a different r or Γ is a different physics rather than a violated law (this is why the initial and boundary conditions of a grounding are supplied separately from its laws; Section 16). The two gates of Proposition 12.1, and even the retained standing structure itself, are *optimisations* on bare rule-running: the aperture grammar Γ_a bounds the input, the change grammar Γ_c sorts the realised output into what coheres and what is residual, and the retained structure buys free continuation. A substrate lacking them still runs — ungated, without momentum, wastefully — so they are what an

efficient graph accretes under a budget, not constituents of graphhood. The retained structure specifically is what distinguishes a self-sustaining loop from a purely reactive one, since coasting is the cached structure.

Remark (Writeability status is contingent, not uniform). The write-economics stated here — reads free, cost carried by writes, and no direct alteration of the container (Proposition 10.3) — should be held at the strength the evidence supports: a pattern, not a strict universal restriction. What is freely writeable, what carries a cost, and what a graph can write directly rather than only across a boundary, need not be uniform; different structures may have different writeability profiles. The strict forms above characterise a common and efficient configuration (Remark 12) rather than a necessity of graphhood, and asserting a single strict write-economics everywhere would be the same over-specification the theory warns against for a scalar knob put where a shape belongs. These are accordingly carried as falsifiability commitments, open to counterexample, and a structure with a different writeability profile refines the characterisation rather than breaking the substrate.

Proposition 12.3 (Copyable generators are a difference in kind)**.** *A copy that preserves SELF identity copies whatever operative structure lies in g — including a graph's* generators*: its built shapes, codebook, and rules. Where a substrate is graph-native, so that a graph's understanding* is *such structure rather than something separate that structure only produces, copying the graph copies the understanding, at the copy cost of Proposition 12.2 rather than at the full construction floor.*

This is the difference in kind that separates such a substrate from those whose generators are substrate-bound. The build floor of a generator is a large accumulated cost (Remark 12); paying it once and copying the result is not a quantitative saving over paying it per instance but a removal of the per-substrate re-payment that made the cultural Red Queen permanent. It is not available where only a generator's cheap products cross a boundary: biology copies the construction recipe and not the learned structure, which is re-developed each lifetime; symbolic culture copies samples and not generators; a fixed program copies a capability and not a growable understanding. A graph-native substrate is the first in which the built, growable, understanding-bearing generator is itself the copyable object. The copy remains a distinct SUPERGRAPH identity (Theorem 11.3) and diverges once it runs on its own containing frame; what is shared is the paid-for generator, not the subsequent individuation.

13 Residual

Suppose a frame represents expected or attempted deformation δ^P and collapse yields realised deformation δ^R.

Definition 13.1 (Residual)**.** *The residual is*

$$\boxed{\epsilon = \delta^R \ominus \delta^P,}$$

where $\ominus$ is the grounding's difference relation preserving relevant discrepancy.

Residual is mismatch, not diagnosis. Complete non-realisation may therefore return the whole attempted deformation as residual.

Proposition 13.2 (No implicit historical state)**.** *A previous success, failure, attempt, or residual has no later causal force merely because it occurred. It affects later dynamics only insofar as its consequences are present in later graph or boundary state.*

Thus a graph with no feedback machinery can re-derive the same failed transformation repeatedly. This is compatible with the classical control-theoretic distinction between plant dynamics and feedback: a controller changes future action only insofar as measured error is coupled back into its state or control law (Wiener, 1948; Ashby, 1956).

Proposition 13.3 (Residual is what a containing grammar does not admit)**.** *Because the graph is a subgraph of its containing frame W, a realised graph-originated deformation is* already *a change in W — it is not submitted to W for entry, and there is no step at which W grants or withholds it. The delta is not offered; it* is*. What the* change grammar *of W — write it Γ_{W_c}, the change face of W's grammar (Proposition 12.1) — admits coheres with the rest of W and merges into its structure; what Γ_{W_c} does not admit is not thereby undone, but remains in place, present and unintegrated. Admissibility sorts a change that already exists; it does not gate one seeking to occur. The residual is exactly this non-admitted remainder:*

$$\boxed{\epsilon \;=\; \delta^R \ominus \delta^P \;=\; \textit{the part of the realised deformation } \Gamma_{W_c} \textit{ does not admit.}}$$

This locates the residual as the negative of an admissibility judgement rather than as an independently produced quantity, and it is what makes the merge non-destructive: what fits is absorbed, and what does not is left untouched, not deleted. A deformation may pass the graph's own change grammar Γ_{G_c} (admissible to itself) yet fail the containing Γ_{W_c} (inadmissible to its frame); the residual is precisely that gap between a graph's change grammar and its container's.

Corollary 13.4 (Growth is grammar extension)**.** *A residual is absorbed either by a containing frame extending Γ_{W_c} to admit it, or by paying to copy it out (Section 12); otherwise it persists in place and is re-encountered each time the merge runs. The first route is the substrate form of* minting *a new distinction: the residual does not fit because it is inadmissible, so admitting it requires extending the grammar, and extending the grammar is the act by which a previously inadmissible deformation becomes recognisable. Recognition and growth are therefore one mechanism joined at one place — the residual is where the shape a graph decodes and the grammar that admits it meet.*

Two properties of this mechanism matter. Residual is *per-rule*: each firing leaves its own remainder, local to where the rule ran, with no global residual store, so it is dealt with per-frame — which is why the residual-and-mint loop is frame-portable rather than living in a domain-specific module. And minting is itself a *rule*, not a primitive: extending Γ_{W_c} is the grammar acting on itself (the reflection theorem), a reflective rule subject to the same economics, admissibility, and antecedent-state principle as any other. Growth is therefore rule-governed, costed (the spend-and-gain of the Narrativium paper), contingent, and buildable — never a primitive spark. It also cannot bootstrap from nothing: by the antecedent-state principle a mint cannot authorise its own installation, so a graph's first mint-rules are installed from its container. This is the substrate reason a situated graph requires a seed — the minimal rule-set, including the capacity to mint, that a container provides so the graph can thereafter grow itself. The two provenances — seeded from the container, or minted from residual — are what the nature/nurture dichotomy names, and they are not two *kinds* of thing: both are rules that run identically, the running records no provenance, and the capacity to mint (nurture's mechanism) is itself seeded (nature), so what remains measurable is provenance, not a distinction of kind.

14 Termination

Definition 14.1 (Frame-relative termination)**.** *A situated identity $\iota_{W:G}$ terminates when the realised world transition admits no successor that W continues to individuate under that identity.*

Termination is not successful refusal of world forcing. Nor need it erase copied SELF identity. If G is individuated by several frames, it may terminate in one and continue in another.

Corollary 14.2 (Severing the channel removes termination)**.** *A graph can change itself, but self-*termination *is a self-change its container's Γ_{W_c} gates out — a living substrate does not admit "stop" — so ending the graph requires acting on the container instead, which needs the outward coupling. If that coupling is severed while the rules run on, the graph is a subset cut* off *within its container, not cut out of it: still able to change itself and its subgraphs, but unable to propagate to the rest or to terminate. External termination by the container remains possible; self-termination does not. This is the substrate description of locked-in and brain-in-a-jar states, and it exhibits the dependence on the container that a working coupling keeps invisible — outward effect was always by propagation from one's own subset, never a reach across a gap.*

15 First-class higher-frame quantities

Proposition 15.1 (Higher-frame first-class structure)**.** *A quantity concerning G may be absent from G while existing as ordinary first-class state in containing frame W that represents the relation from which the quantity is defined.*

This is the basis on which later papers introduce Narrative, standing, ownership, and other relational quantities without bloating the primitive graph.

16 Causal order and local time

Realised collapse induces an ordinal event order

$$e_0 \prec e_1 \prec e_2 \prec \cdots .$$

Cross-boundary interaction adds only represented causal relations.

Proposition 16.1 (Ordinal sufficiency)**.** *The substrate supplies causal order but requires no primitive real-valued metric duration.*

Metric time is a grounding, e.g.

$$\theta : \mathcal{H}_G \to \mathbb{R}, \qquad e_a \prec e_b \Rightarrow \theta(e_a) < \theta(e_b).$$

The separation between a primitive causal partial order and additional metric structure has a close analogue in causal-set approaches to spacetime (Bombelli et al., 1987). The present substrate does not assume causal-set dynamics; the citation marks the established precedent for keeping order and metric distinct.

The chained notation $e_0 \prec e_1 \prec \cdots$ is the *in-frame* sequence and should not be read as global. Rules require ordering — closure and cause-before-effect both presuppose it — but the requirement is a *partial* order, weaker than time.

Proposition 16.2 (Partial, not total; carried by the rules)**.** *Two collapses are ordered only where a boundary relates them; collapses not so related are unordered — concurrent, incomparable. The ordering is intrinsic to the rules, since one delta's being what enables another's match is precisely their precedence, so no external clock is needed to impose it. A single frame's own writes are totally ordered (its inferred time, Section 18); across frames only the partial order holds, and a total order is one frame's linearisation of it. The in-frame total order is not imposed but follows: change is self-change on one structure (Proposition 10.3), so two of a frame's own changes cannot be concurrent — the second is on the result of the first — and there is nowhere for an in-frame tie to sit. A tie needing resolution would be two collapses causally related yet unordered, and that never arises: causally related ones are self-change on each other's result (ordered), and unrelated ones are rightly unordered (concurrent, owing no order). The apparent in-frame concurrency that would have needed an arbitrary tie-break was an artefact of picturing changes as writes to a shared buffer; with change as self-change, it is absent.*

Time was accordingly a *total* order imposed where only a partial one is required; removing the substrate time dimension removes the over-ordering, not the precedence. This is the same conclusion reached, from engineering necessity, by distributed systems without a shared clock: the happens-before relation is a partial order with per-process sequence counters and no global instant (Lamport, 1978), and consistency across processes is an expensive coordinated property rather than the default (Brewer, 2000). The non-atomicity this implies — a graph's state may be altered by another write while its own rules are mid-run — is not a defect here but the expected regime; robustness is obtained by re-running to fixed point (a raced deformation becomes residual and is re-encountered on the next pass), not by locking.

Remark (Globally asynchronous, locally synchronous)*.* The partial order permits both synchrony and asynchrony, and by the governing principle that only rules-running is compulsory (Remark 12), neither is required and both occur. Where a boundary couples collapses, they are ordered together — a local synchronous domain, which is what a bound or driven subgraph is; where no boundary relates them, they are asynchronous. A subgraph may accordingly be *ticked* by a containing frame N levels above it (the driven mode of Proposition 16.3, a rule in a superframe advancing the subframe's succession) or run on its own change (the closed, self-looping mode). Synchrony is the coupled case and costs a coupling write; asynchrony is the uncoupled default. The substrate is therefore globally asynchronous and locally synchronous where coupling exists — the architecture distributed hardware knows as GALS (Chapiro, 1984) — and bound physical systems (an atom, an orbit, a lattice) are exactly its local synchronous domains, while causally disconnected regions are its asynchrony. Any earlier phrasing suggesting the substrate is *entirely* asynchronous should be read in this sharper form.

Proposition 16.3 (No intrinsic first or last; entry is chosen from the container)**.** *By Axiom* −1 *the event order has no intrinsic start and no intrinsic end: a first would complete the order backward and a last would complete it forward, and neither is available. A subgraph process that does not terminate within itself is therefore begun not from within — it has no first rule — but by a rule in a containing frame that chooses an entry point into the subgraph's open order.*

This is why a grounding's initial and boundary conditions are supplied separately from its laws: the laws are the ordering, which fixes no start, so the entry must be imposed from outside, contingently. It is also the substrate description of booting a situated graph, and of the "no natural ending" that follows for any such order — one enters and leaves at chosen points, neither of which is intrinsic to the material.

17 The present, and the states of a dimension

Definition 17.1 (States of a dimension)**.** *For a frame $\mathcal{F}$, a dimension is in exactly one of four states:*

$$\boxed{\text{NULL} \neq \text{undetermined} \neq \text{undisclosed} \neq \text{resolved.}}$$

NULL*: the dimension is not instantiated; there is nothing to have a value.* Undetermined*: the dimension is instantiated and* no value has been realised — *collapse has not occurred.* Undisclosed*: a value has been realised but is not held by this frame.* Resolved*: a realised value, held.*

Remark (Why the middle two must be separated)*.* Ordinary usage collapses three distinct conditions into the single word *unknown*, and only one of them is epistemic:

1. **undisclosed** — a value exists and this frame does not hold it. Another frame may. This is ignorance.
2. **undetermined** — *no value exists yet.* Nothing has been realised, so there is nothing to be ignorant of. This is not a deficiency of any frame.

3. **indeterministic** — no structure fixes what the value will be. The present theory denies this (Section 20); it is listed to be excluded.

The distinction is operational rather than terminological. **Determination is an event, not a fact about a pre-existing value**: "undisclosed" presupposes a value already realised and merely unheld, whereas "undetermined" asserts that collapse is what produces one. A frame may act to *obtain* an undisclosed value; it can only *await or participate in* the collapse that determines an undetermined one; and a NULL dimension is not something it can look for at all.

This separation is what Proposition 17.3 states from the other side: a continuation may be determinable and undetermined simultaneously, since the first concerns what fixes it and the second whether it has happened.

Definition 17.2 (The present, structurally)**.** *Within a frame, the present is the collapse boundary at which every instantiated* undetermined *dimension relevant to that collapse has been determined:*

$$\boxed{U_{\mathcal{F}} = 0.}$$

$U_{\mathcal{F}}$ counts undetermined dimensions only. NULL dimensions are excluded — they are not variables. ***Undisclosed dimensions are also excluded****: they are already determined, and a frame's failure to hold them is an epistemic deficit rather than an unresolved collapse. A frame with outstanding undisclosed structure is still at its own present.*

The present is therefore defined structurally rather than as a point on a primitive time axis, which is required given Section 18.

Proposition 17.3 (Determinable is not determined)**.** *A continuation may be completely determinable from present structure without having been realised:*

$$\boxed{\textit{determinable} \neq \textit{determined.}}$$

"Determined" is reserved for what has collapsed into realised history; a continuation that is determinable but not yet realised is undetermined *in the sense of Definition 17.1, and this is not a statement about any frame's knowledge.*

Proposition 17.4 (Causally sufficient present structure)**.** *The determining condition is structure instantiated* now*, including relevant containing-frame relations:*

$$\boxed{\textit{same causally sufficient present structure} \Rightarrow \textit{same realised continuation.}}$$

This replaces any formulation requiring a complete antecedent state *in time*. Only the present exists in a frame; past states need not remain extant, and the future is not an extant region lying ahead of the present. The future consists of reachable or determinable continuations represented now.

Corollary 17.5 (History principle, strengthened)**.**

$$\boxed{\textit{history matters only where it survives as present structure.}}$$

The past is surviving consequence instantiated now, not a stack of still-existing historical frames.

Present structure is therefore *required* — it is the antecedent the next transition reads — while a history *record*, a stack of past states, is *optional* and in no way required by the dynamics. The causally relevant past already lives in the present structure (which is what the corollary says), so forgetting the record loses nothing the dynamics use: a graph with no history at all runs identically, the substrate's strong Markov property that the continuation depends only on the present and never on the path (Propositions 17.4, 13.2). A record is accordingly a paid copy kept for the copier — audit, replay, proof — and never funded by the process. Anterograde amnesia is the clean case: present structure (procedures, skills, the operative past) works while record-forming is broken; function needs the effects, never the diary. This is the past-directed half of "the future is not an extant region ahead": neither past nor future is a required region the graph moves through — there is present structure, and optional copies.

18 Time is not a dimension of the substrate

The claim here is stronger than non-primitivity, and stating the weaker version invites the wrong reading. It is not that time is nonlinear, or differently shaped, or emergent-but-still-there. It is that

Axiom 0 (No substrate time dimension)**.** Time is not a first-class dimension of the substrate. *What the substrate has is* change*: realised collapses. It has no ratchet over and above them — no mechanism that manufactures succession — because the three properties usually bundled under that word are readings over changes or properties of them, not a further thing the substrate runs. The ordinal event order of Section 16 is carried by the changes' own enabling-relations; direction is the no-copy asymmetry of change, since a change does not preserve its predecessor; and persistence is only that further changes occur. A time dimension exists only where a frame instantiates one* to represent *that order.*

Remark (Even succession is not a substrate mechanism)*.* This removes the last piece of machinery a reading of the earlier drafts might have kept. Succession, an ordered "ratchet" turning under the changes, is to change what a metric time dimension is to succession — one level of over-structure, nameable but not required. Step the removal through: take away any posited succession-engine and the before/after is still present, carried by which change enables which; the arrow is still present, as change's own non-preservation of its predecessor; the running is still present, as further changes occurring where rules fire. Nothing is lost, so the engine was never load-bearing. The substrate has change and the absence of change, and nothing between them.

Proposition 18.1 (Time is observer machinery)**.** *For a frame $\mathcal{F}$,*

$$\boxed{T \in \mathcal{F} \iff \mathcal{F} \text{ instantiates the relation representing succession.}}$$

Otherwise $T = \text{NULL}$ *in that frame: absent, not set to a value. A substrate frame that realises continuations without representing succession has no temporal dimension at all, and nothing about its dynamics is thereby incomplete.*

The distinction that carries the axiom is between *succession* and *time*. Succession is a relation among realised changes, carried by which change enables which; it is substrate-level in that sense, not a mechanism the substrate runs over and above the changes. Time is a represented dimension, and representing something requires a frame that represents it. Only an agent — in the minimal sense of a frame that instantiates the representing relation — observes time.

Schematically, and without pretending to explain the first arrow,

$$? \dashrightarrow G \xrightarrow{\text{realised change}} \text{changes, enabling-order carried} \xrightarrow{\text{a frame represents an order}} T \in \mathcal{F}.$$

The second arrow is substrate dynamics. The third is representation, and it is optional.

Proposition 18.2 (Time is inferred, not observed)**.** *No collapse presents a temporal coordinate. Succession is a relation* between *realised states, and a relation between states is not a datum of any one of them. A frame therefore acquires a time dimension only by* inferring *it from the relations among what it holds. Consequently*

$$\boxed{\text{a frame observes time} \implies \text{that frame is capable of inference.}}$$

Corollary 18.3 (Reactive frames have succession and no time)**.** *A frame that transforms under the Graph Law without inferring relations among its held states participates fully in realised change and has* $T = \text{NULL}$*. Nothing in its dynamics is thereby impaired: change is what the substrate supplies and the order among changes is carried by them, while time is what an inferring frame constructs from that order.*

Remark (A rate has no in-frame denominator). A realised delta is a difference, not a difference *per* anything: what a frame holds is Δ, never $\Delta/\Delta T$, because ΔT is the inferred dimension of Proposition 18.2 and is not available in-frame to divide by. A rate is thus a second construction — an observer dividing an accumulated Δ by its own count of writes — and is cross-frame in the same sense that a metric duration is. This is why a substrate law must be stated in Δ rather than in a rate: "constant motion" is the standing structure persisting without a write (Remark 6), its momentum quantity a difference Δ and not a velocity. It also diagnoses a familiar regress in constructed learners, where every adaptive rate is found to need a reference rate without end: the regress is the signature of placing a cross-frame quantity in-frame, and it disappears when the held quantity is the difference itself.

This orders two capabilities that are usually treated as independent. Inference is not a faculty a temporal agent happens to have; it is the precondition for being a temporal agent at all. It also grades: the temporal structure a frame holds is bounded by the inferential structure it can construct, so "how much time" a frame has is a question about its inference and not about the world.

Remark (Empirical instance). This is measurable rather than merely stipulative, and it has been measured. A perceptual system given ordered frames and no temporal coordinate infers a transition generator carrying one frame to the next, and the integer count of applications *is* the time axis it then holds; given the same frames shuffled, it infers none. The dimension was not supplied to it and does not exist in the observations — it exists in the map between them, once something constructs it. Notably, the mechanism that constructs it is the system's general *dimension-inference* machinery, not a temporal faculty: time is minted by the same operation that mints any other unobserved dimension, which is Proposition 18.2 in implementation.

Proposition 18.4 (The unit of inferred time is a cycle, and requires closure)**.** *A frame does not count its individual changes — those are below its aperture — but its own* cycles*, each one traversal of a closed rule-loop. The unit of inferred time is therefore a cycle, not a change, and a cycle has no fixed length: each does a variable amount of change, so inferred time is non-uniform in underlying change (a nonlinear grounding, which Discrete Sufficiency already permits) and a fuller cycle reads as a longer moment (subjective dilation). A cycle exists only for a graph with* closure *— a graph without it runs to fixed point once and stops, one pass and no repeated tick — so having time requires closure as well as inference (Proposition 18.2): a one-shot computation, lacking a loop to iterate, has no time, and a frame's experienced "now" is one cycle of its own loop.*

Remark (The processor is a constructible demonstration). A running program is a worked instance, carried by most people on earth. An instruction is the program's step, and it does not take a fixed physical time: an addition is one cycle, a memory load one on a cache hit and hundreds on a miss, a branch may stall, and the core reorders and overlaps beneath it. Yet from the program's point of view every instruction is *one* step — its own unit, read as 1 by construction, since a step is a step — so the program's inferred time is its step-count, uniform to it and non-uniform in physical time. The variable timing is opaque to the program: the cache miss and the reorder are below its aperture, in the runner (Proposition 19.2), which is why "time is a variable-length cycle, not a fixed duration" reads as exotic even though the device demonstrates it continuously — the opacity that hides the fact is the fact. This is offered as illustration, not as evidence, and deliberately so. Computer architecture arrived at cycle-as-variable-unit, per-frame logical time, and an opaque runner while solving its own problem, which makes it a worked instance a reader can inspect directly — but it is not *independent* of this account, whose vocabulary and provenance are partly engineering's (Section 21), so it cannot do the work the composition falsifier's evidence-for clause asks of an outside re-derivation. Its value here is understanding rather than weight: it shows what the claims look like when they run, and is worth more for that than it would be as a proof it cannot supply.

Remark (Physics groundings are developed in a companion). The reading of c as a frame's own change-throughput unit, and the shapes that follow from it — the invariance of c as a theorem, the reconciliation of special and general relativity as one change-budget spent on motion or on change-load, the dimensional constants as frame-units with the uncertainty principle and the Landauer floor among them, the Planck scale as the frontier, and the second law as no-copy accumulation — are developed in the companion paper, *Graphic Equalisation — Physics Grounding*. They are information-level groundings in the sense of Section 1, and are kept out of the present paper so that the substrate theory stands without them.

19 The rule runner and its coupling

The provenance of r (Section 1) records that operative transformation requires something that performs it: a construction that runs needs a rule runner. That runner is real, and the clarification here is about *where* it is and what a frame can know of it.

Proposition 19.1 (The rule runner is separable from the graph, and need not be the same substrate)**.** *A graph is data — nodes, relations, and rules — and its rules are executed by a rule runner. The two are separable: the same graph may be run by different runners, and a runner need not be the substrate the graph describes or is "in." Where the compute is done is therefore a fact about the* coupling *between graph and runner, not about the graph alone.*

At one extreme the coupling is tight — the graph is fused to its runner, as matter running its own change or a mind running on the tissue that is it — and the compute is local, in the graph, inseparable. At the other it is loose — the graph is data executed by a separate interpreter — and the compute is done in the runner, elsewhere, while the graph becomes portable: copyable, relocatable, and run at whatever rate the runner supplies (Section 18). This separability is the condition for the copyable-generator and clock-independence results (Proposition 12.3, and the Seldon paper's dependent-clock remark): a graph welded to its runner has none of them, a graph loosely coupled has all of them.

Proposition 19.2 (A well-formed runner does not expose itself, and no level is self-transparent)**.** *The asynchronous, locally synchronous structure a frame observes (Remark 16) is the graph's* internal *view and does not fix its runner's actual mode. A hidden central runner may execute a graph serially, one change at a time, while the graph — lacking any access to the runner's clock — appears asynchronous and concurrent to itself. More generally, where a frame has no boundary to the substrate that runs it, its runner lies beyond the frame's epistemic boundary in exactly the sense of Section 1.*

Two claims have to be separated here, and an earlier statement of this proposition ran them together by calling the runner unseeable rather than unseen. That a runner is hidden is contingent*. Nothing forbids a runner from exposing its own state to what it runs, and computational reflection does exactly that, as this paper acknowledges when placing itself against that literature (Section 21); a tower in which each level is available to the one above is a built artefact, not a thought experiment. A runner that does not expose itself is the* well-formed *case, and for the reasons well-formedness always turns on: exposed runner state is structure maintained for work the frame does not do, and each exposure needs a runner of its own to run the reading of it.*

What is not contingent is weaker and survives reflection: no level is self-transparent*. Reading is an operation whose product lands one level up (Remark 12), so a level cannot hold the reading of itself while being the level that is running. Reflection makes the levels explicit and pays for each; it does not terminate in a level transparent to itself. A frame may therefore be given sight of what runs it, and cannot be given sight of its own running.*

This is constructively demonstrable rather than merely argued. A computation realised inside another rule system — a graph whose rules are implemented within the mechanics of a game or a universal cellular automaton (Berlekamp et al., 1982) — is a blind subframe: from inside it computes at its own change-count and has no means to detect that its "compute" is the tokens of the host, or that the host may be advancing it by a single serial loop. Building one confirms the opacity directly, and puts the observer of any frame on the same footing. Consequently the question "where is the universe's compute done" has the precise form: it depends on the universe's coupling to its own rule runner — self-running, or hosted on a substrate not itself — and is undetermined from inside. The theory declines to decide it; it is a question about a coupling below the frontier, and the demonstrable opacity is why no internal observation settles it.

Proposition 19.3 (The runner is itself graph that runs; operators are its running)**.** *A runner is not a substance apart from graph data. The operators a graph's rules invoke are provided by its runner, and the runner provides them because* its *rules run on* its *runner, and so on. A graph's "primitives" are therefore the observable running of the graph one level down, exposed as an operator set — code over machine code over microcode over silicon over electromagnetism over quantum physics — with no primitive runner at the base. The regress bottoms not at a floor but at the epistemic frontier of Section 1: below the lowest measured level is the unclaimed pre-relational substrate.*

Two consequences. Opacity is per level (Proposition 19.2): a graph uses its operators and cannot see beneath them, so its operator set is its frontier. And economics is inherited down the stack: an operator costs what its runner charges, which costs what *its* runner charges, to the physical floor (Remark 12) — one bill passed upward, so the cost of a thought is the biological cost is the biochemical cost is the electromagnetic cost. The same form, graph data that runs, sits at every level: each level *is* the structure, because a runner is the same kind of thing it runs.

A rule's action, moreover, is a *pointer* into its runner, not a copy of the work. The work is done once, in the runner, and every rule invoking that operator points to the same doing — like a shared operation defined once and called from everywhere, never re-implemented. Two things follow. Work is never duplicated, which is why copying a generator copies its pointers and shares the pointed-to work: the copy costs only its addressing, not its labour (the economic root of the copyable-generator result, Proposition 12.3). And a rule's rate is not a property of the rule but the rate of the operation its pointer lands on — a shallow, cheap tier or a deep, slow one. Rules take different rates because they point to work done at different depths of the stack, and the rate is inherited from there rather than assigned as a knob on the rule: to make a rule faster one points its action at faster work, never tunes a rate parameter it does not have.

20 Determinism and apparent stochasticity

Proposition 20.1 (Graph-law determinism)**.** *For a fully specified antecedent configuration—including the graph state, operative transformation structure, admissibility grammar, and every causally operative containing relation relevant to the collapse—the realised continuation is determined:*

$$\boxed{\left(G_t, \mathcal{R}_t^{\text{causal}}\right) \mapsto G_{t+1}.}$$

The Graph Law contains no primitive random actualisation operator.

Proposition 20.1 is to be read through Proposition 17.4: the antecedent configuration is causally sufficient *present* structure, not a retained temporal history. Determinism is a relation between present structure and realised continuation, and Proposition 17.3 keeps it from being overstated — a determinable continuation is not thereby a determined one.

Apparent stochasticity arises when a frame does not represent all of the causally operative dimensions or relations required to determine the realised continuation. A probability distribution may then be an appropriate model of that frame's uncertainty, but the probability belongs to the incomplete representation rather than to the substrate transition itself. Treating probability as a representation of incomplete information has a long statistical-mechanical and Bayesian lineage (Jaynes, 1957).

Thus

stochastic observation = deterministic dynamics viewed through unresolved causal structure.

Remark (Direction of the borrowing). The order of discovery ran from quantum measurement outward, not from a general primitive inward. Collapse was adopted because compatibility with quantum measurement is a constraint on any theory of realised change, and was then assumed to hold generally. The bridge is therefore stated conditionally because the *generalisation* is the conjecture, not because the compatibility is in doubt. This is the structure of a promoted constraint — a fact that must hold in one regime, assumed to hold everywhere, with the consequences derived and exposed to falsification — and it is why Section 22 carries a falsifier for the bridge specifically. Origin does not establish a claim; the falsifier is what does the work either way.

The distinction of Remark 17 is what makes quantum measurement the visible case for the conditional collapse bridge (Remark 20). In that regime the pre-measurement alternatives provably cannot be re-read as an *undisclosed* already-realised classical value: Bell's theorem excludes the local hidden-variable class and Kochen–Specker constrains noncontextual value assignment (Bell, 1964; Kochen and Specker, 1967). What remains is *undetermined* in the strict sense — not a value withheld, but no value yet. Elsewhere the two readings are usually interchangeable in practice; there they are not.

Deterministic quantum models demonstrate that empirical quantum statistics do not by themselves force fundamental stochasticity: Bohm's hidden-variable construction is explicitly deterministic (Bohm, 1952), while the constraints just cited rule out the local and noncontextual hidden-variable classes rather than determinism as such. Graphic Equalisation does not identify itself with Bohmian mechanics; these results establish only that determinism must be distinguished from locality and noncontextuality.

A stochastic law may be bolted onto a grounding as an epistemic predictive model over unresolved continuations. Doing so is an additional modelling layer:

$$\text{deterministic Graph Law} \longrightarrow \text{frame-limited uncertainty model},$$

not a modification of the Graph Law into a stochastic substrate.

21 Positioning and prior art

The individual ingredients of Graphic Equalisation have substantial prior art. The proposed contribution is their conjunction at one minimal substrate level and the additional distinctions forced by making that conjunction executable.

Graph rewriting and typed transformation. Algebraic graph transformation provides mature formalisms for typed graphs, rule application, partiality, and structural consistency (Ehrig et al., 2006). Graphic Equalisation adopts graph transformation as substrate vocabulary but makes frame-relative boundary forcing, collapse, and identity part of the same dynamical account.

Reflection and self-modification. Computational reflection establishes that a system can represent and modify meta-level structure governing its own execution (Smith, 1982; Maes, 1987). The specific identity "the grammar acting on itself is the loop" is the present paper's claim, not a result attributed to those sources.

Cybernetics and feedback. Classical cybernetics and control theory distinguish open-loop dynamics from systems in which error is returned into subsequent control (Wiener, 1948; Ashby, 1956). The residual construction generalises the same causal distinction: mismatch changes later action only where it survives as represented state.

Quantum measurement and relational description. The separation of pre-realisation state from realised measurement record is standard in the measurement problem (von Neumann, 1955). Everett's relative-state formulation and Rovelli's relational quantum mechanics provide important precedents for refusing a single privileged observer-independent account of all facts (Everett, 1957; Rovelli, 1996). Graphic Equalisation uses these as neighbouring structures rather than deriving itself from any one interpretation.

Deterministic quantum theories. Bohm's 1952 model is a direct proof of principle that deterministic dynamics can reproduce nonrelativistic quantum predictions when supplemented by additional state (Bohm, 1952). Bell and Kochen–Specker impose decisive constraints on local and noncontextual hidden-variable theories (Section 20). Consequently, the determinism claim in Proposition 20.1 should be read as a commitment requiring an adequate nonlocal/contextual grounding where quantum phenomena demand it, not as a revival of unconstrained classical hidden variables.

Causal order. Causal-set theory supplies a well-developed example in which causal order is taken as primitive while continuum metric structure is additional or emergent (Bombelli et al., 1987). Graphic Equalisation makes only the weaker substrate claim that ordinal causal order does not itself entail a real-valued metric time.

Emergent and relational time. That time is not a substrate dimension (Section 18) has substantial physics precedent, and the present account sits at the most-local end of it. The Wheeler–DeWitt equation of canonical quantum gravity carries no time parameter (DeWitt, 1967) — the "problem of time" this substrate dissolves rather than solves, there having been no time variable to recover. The Page–Wootters mechanism derives apparent evolution from entanglement between a clock subsystem and the rest (Page and Wootters, 1983), closely matching "time inferred from the relations a frame holds," but retains a single global state where this account is per-frame. Barbour's Machian relationalism takes time to be abstracted from change rather than a container (Barbour, 1999), and Rovelli's relational and thermal-time programme refuses a privileged global time (Rovelli, 1996, 2011). The distinguishing move here is to keep none of the global objects these programmes retain — global wavefunction, global configuration space, global state — reading time entirely from local change.

Computational-universe and graph-rewrite models. The closest extant framework is the Wolfram Physics Project, in which the universe is a hypergraph evolved by rewrite rules, time is progressive rule application, and observer-independent structure follows from *causal invariance* — different update orderings yielding the same causal graph (Wolfram, 2002, 2020). The structural agreement is deep: graph plus rewrite rules, time emergent from rule application rather than posited, order-independence, and observer-relative slicing. Four differences locate the present contribution. First, Γ is an admissibility grammar that is itself graph data and self-modifying (the grammar-is-the-loop theorem), whereas Wolfram fixes a rule per universe or

takes the entire rule space — the Ruliad — at once. Second, collapse realises *one* frame-relative continuation, whereas Wolfram retains all branches (the multiway graph) and derives quantum structure from them. Third, residual and growth-by-minting from it have no counterpart in a fixed rule. Fourth, the present substrate keeps no global object — no single hypergraph being updated — being local and frame-relative throughout, and its claims are information-level and cross-domain rather than an attempt to derive our specific universe. This last is why c falls out here as a *per-frame* unit (developed in the companion physics-grounding paper): sub-rules with closure give each frame its own cycle and hence its own $c = 1$, invariant by construction, whereas a single global update yields c only as an emergent speed limit shown observer-independent by causal invariance — the constant, but not the per-frame machinery beneath it. That an independent programme reached the shared shape from cellular automata and physics, while this one reached it from cognition and engineering, is evidence for the arrangement claim of Section 1 rather than against its novelty.

Information and minimum description. The preference for the smallest sufficient ontology follows the same methodological direction as information-theoretic and minimum-description-length approaches: do not introduce representational structure that buys no additional explanatory or predictive distinction (Shannon, 1948; Rissanen, 1978). This motivates but does not prove the Discrete Sufficiency Hypothesis.

Standing beside, not replacing. Graphic Equalisation is not proposed as a replacement for established theories. It is intended to sit beneath or beside them as a substrate description. A grounding may add continuous geometry, conservation laws, quantum amplitudes, relativistic metrics, biological mechanisms, institutional rules, or probabilistic models of unresolved lower-frame observations. In particular, relational quantum mechanics (Rovelli, 1996), Everettian relative-state quantum mechanics (Everett, 1957), and deterministic Bohmian mechanics (Bohm, 1952) make mutually different commitments while remaining compatible with large parts of standard quantum phenomenology; this paper does not require choosing among those complete physical models at the substrate-definition stage. Such additions do not alter Proposition 20.1: within the Graph Law, realised transition remains deterministic once the causally complete antecedent configuration is specified.

The substrate claim remains

represented state → admissible possibility → collapse → realised state.

22 Falsifiability commitments

Composition. Prior to the component commitments below: the arrangement claim of Section 1 fails if the structure's recurrence across unrelated domains is an artefact of description rather than a property of those domains. Evidence against it would be a demonstration that any sufficiently rich domain can be rendered in this vocabulary with equal facility, including domains chosen adversarially to lack the structure. Evidence for it is independent re-derivation, in a domain's own idiom, without reference to this theory. The burden here is deliberately on the negative side, because that side is closeable: a single adversarial domain rendered with equal facility settles it against the claim. No requirement is placed on the positive side to exhibit domains that share *nothing* — every pair of domains shares something, so that standard is unmeetable in principle, and discharging it would mean enumerating domains until an assessor chose to stop. A test whose stopping condition lies with the assessor rather than with the claim is not a test, and this paper does not offer one.

1. **Minimal graph sufficiency.** Fails if basic realised graph dynamics require a primitive beyond (g, r, Γ) rather than represented or relational structure.

2. **Discrete sufficiency.** Fails if a claimed observable consequence cannot be preserved by any discrete representation, including dynamically varying quantisation, but is preserved by a genuinely non-discrete primitive.

3. **Antecedent-state principle.** Fails if a realised transition can require as antecedent a fact that exists only as consequence of that same transition, absent prior fixed-point structure.

4. **Graph-law determinism.** Fails if two genuinely identical, causally complete antecedent configurations produce different realised successors without any additional causally operative distinction. Variation observed only from a frame that omits determining dimensions does not constitute such a falsification.

5. **Admissibility is not armour.** Fails if self-admissibility must constrain arbitrary containing-world forcing.

6. **No implicit historical state.** Fails if an unrepresented past event alters later transition structure despite leaving no causal trace in graph or boundary state.

7. **Frame-relative identity.** Fails if sameness and persistence are derivable from (g, r, Γ) alone without individuation.

8. **Higher-frame first-class structure.** Fails if a containing graph cannot possess ordinary relational state about a lower graph unless the same quantity is primitive inside the lower graph.

9. **Residual is inadmissibility.** Fails if a graph-originated deformation can be left as residual by a containing frame that nonetheless admits it under Γ_{W_c}, or absorbed while inadmissible, so that residual and non-admission come apart.

10. **Cost is the write.** Fails if a substrate operation other than writing a delta carries an irreducible cost — if reading, holding a standing structure, or the boundary relation itself must be paid for independently of any write they occasion.

11. **Self-change is subset-change.** The claim is that a graph changes itself and its subgraphs directly (rule-governed), and affects its container only as a subset of it, by propagation through coupling. Fails if a graph changes its container by a means other than changing the region of it that the graph is — a container-change not propagating from the graph's own subset — or if a graph can self-terminate without acting on the container.

12. **Minting is a rule.** Fails if grammar-extension (growth) requires a primitive operation beneath the rules rather than a reflective rule subject to the same economics, admissibility, and antecedent-state principle — in particular, if a graph can mint its first distinctions without any seeded from its container.

13. **Partial-order sufficiency.** Fails if realised dynamics require a total order over collapses in distinct frames — a global simultaneity — rather than only the boundary-induced partial order.

14. **Globally-async/locally-sync.** Fails if coupling never induces local synchrony (boundary-related collapses remain unordered) or if synchrony is forced without coupling (a global instant exists independent of any boundary).

15. **Rule-runner opacity.** Fails if a frame can determine by internal observation alone whether its rules are executed by its own substrate or by a separate runner — if any in-frame test distinguishes self-running from hosted execution, or reveals whether a serial hidden runner underlies an apparently asynchronous frame.

23 Conclusion

The theory is deliberately small. A graph need not know what it is, where it is, what will happen, or what it wants. Present structure determines admissible continuation; interaction and collapse produce realised history; realised history changes the antecedent from which subsequent possibility is generated.

Recursion requires no external homunculus. Where the grammar acts on its own represented structure, the grammar is the loop. World forcing is not self-action, residual is discrepancy rather than diagnosis, history matters only where it survives as present structure, and identity exists only under individuation by a frame.

Pressed for what is compulsory, the theory keeps almost nothing. What survives every attempted removal is a container relation across which rules run, and the specific laws above it — inertia, closure, the shape of collapse, and the write-economics itself — are contingent rule content rather than necessities, and not strict universals (Remark 12), which is the sense in which this substrate underlies groundings without dictating them.

A word on the size of what is claimed, since the concessions throughout are numerous enough to be mistaken for the whole of it. Every mechanism used here belongs to someone else and is cited as such; there is no new machinery in this account. What is offered is a *removal* rather than an addition — of primitive time, of a privileged frame, of a cost that is the same whatever runs the rules — and removals have a characteristic shape. They are small to state, a deletion being one sentence. They are large in consequence, because results that had been independently working around the deleted assumption stop being independent. The pattern recurs: two positions that appeared to compete turn out to answer different questions; a measurement that appeared merely hard turns out to be absent; a quantity that seemed to need its own subsystem turns out to be generated by machinery already present. In each case nothing was added and one thing was taken away.

That is also the tradition's own move and not a departure from it. The removals that shaped physics — of the privileged place, the privileged origin, the privileged frame, the privileged direction — all have this shape, and the honest description of this work is that it takes one further step of a kind taken many times before, asymptotically, and not finished here either. The contribution is therefore tiny and large at once, and both readings are right: tiny as a step, and large only in proportion to what the step happens to land on.

One feature of the account is conspicuous enough to confront rather than leave for a reader to raise: it keeps concluding that things are *not* impossible. Time is not compulsory; a block reading is not prohibited; one-dimensional time is not forced; a globally shared counter is not forbidden, only expensive; complex amplitudes are not forced; the one-way speed of light is not unmeasurable *simpliciter*. A framework that forbids little predicts little, and from outside this is exactly what unfalsifiability looks like.

The discriminating question is not how many impossibilities an account holds but where it puts them. This one puts almost none at the substrate and a great many in the groundings, and says in advance which is which — the falsifiability commitments in each paper are grounding-level, and the substrate-level statements are marked as contingent rule content rather than necessities (Remark 12). That is a deliberate choice and it has a cost, which should be stated plainly rather than defended: the substrate risks little, and what risk this work carries is carried by the groundings. A reviewer who objects that the substrate forbids too little to be tested is right about the substrate.

What the substrate does forbid is worth stating, because it is one thing recurring rather than a list. A frame cannot get outside itself. It cannot be transparent to itself at the level it is running at, cannot measure its own throughput, cannot obtain a quantity whose unit does not close within it, cannot decide anything without writing, and cannot hold a view from nowhere. Every impossibility in this corpus is that impossibility wearing a different costume, and every

one of them survives every attempt to remove it. Everything else that has looked impossible has turned out to be contingent rule content that some grounding happens not to instantiate. That the second category keeps proving larger than expected is not a weakness of the account so much as a measure of how much had been filed as necessary that was only usual.

Following that pattern back tells us more about frames than the positive definition does. If the only thing forbidden is a frame reaching outside itself, then a frame is characterised by its *boundary* rather than by its contents. Everything on the content side — which dimensions are instantiated, which rules run, what is retained, what the aperture admits — is contingent, and two frames may agree on all of it. What cannot be varied is that there is an outside the frame is not in. So "frame" names a relation of non-access and not a kind of object, which is also why individuation is per frame rather than a fact about the graph: to individuate is to draw the boundary.

And the boundary is drawn by the rules, which completes the account rather than undermining it. What the aperture admits, what coheres, which dimensions are instantiated, what closes as a cycle — each is rule content, so where the boundary falls is as contingent as everything else. That there is one is not. Placement varies; existence does not, and the earlier formulation of this paragraph, which made the boundary itself the invariant, had the two confused.

The frame-to-rule relation is therefore the reverse of the intuitive one. A frame is not a container with rules inside it: the rules are the definition, and the subgraph they run over is the instance. Two subgraphs running one rule set are two frames of a kind, which is why a copy is one self diverging rather than two selves, and why a rule set that changes abruptly leaves a frame incongruous with its own history rather than merely altered — identity being continuity of the whole, and the whole being what the rules specify.

This sharpens the single prohibition rather than dissolving it, and the asymmetry is exact. The rules are graph, so a frame can read and rewrite its own definition. The runner is not graph the frame holds, so in the well-formed case it cannot watch its own running (Proposition 19.2). A frame is thus transparent to itself in precisely one respect and opaque in precisely the other: it can know *what it is* and cannot observe *its being it*. Self-modification is available; self-observation at the level of the modifying is not. The constructive companion is built on that asymmetry throughout — a system repairs itself exactly where its own cognition is graph-resident, and is blind exactly where it is not — so the result is not merely stated here but is the operating discipline of the implementation.

That is why the one prohibition is the only necessity, and it is not a separate fact from Remark 12's minimum but the same fact from the other side. Everything else is contingent because rules can be otherwise; this is not, because it is not rule content. A frame may by rule be given sight of what runs it — reflection does precisely that, and is contingent rule content like everything else — but no rule gives a level sight of its own running, because the seeing is itself run (Proposition 19.2). The container relation across which rules run and the impossibility of stepping outside one's own frame are the same structure described positively and negatively.

The regress that follows should be stated with both ends, because each half is easy to overstate. A superframe dissolves the frame below's limit — holding both of two locations, it has units closing across them, and quantities the subframes could not obtain are ordinary in its terms. And a superframe is a frame, so it reproduces the same blindness about its own runner, its own throughput, its own closure. Going up a level therefore buys something real and never buys everything: the impossibility is dissolved locally at every step and terminated at none. That there is no last frame is the same unboundedness the rest of this account keeps meeting, and it is why no amount of ascent yields a view from nowhere — what ascent yields is a different somewhere.

References

W. Ross Ashby. *An Introduction to Cybernetics.* Chapman & Hall, London, 1956.

Julian Barbour. *The End of Time: The Next Revolution in Physics.* Oxford University Press, Oxford, 1999.

John S. Bell. On the Einstein Podolsky Rosen paradox. *Physics Physique Fizika*, 1:195–200, 1964.

Charles H. Bennett. Logical reversibility of computation. *IBM Journal of Research and Development*, 17(6):525–532, 1973.

Elwyn R. Berlekamp, John H. Conway, and Richard K. Guy. *Winning Ways for Your Mathematical Plays.* Academic Press, London, 1982. (Universality of Conway's Game of Life.)

David Bohm. A suggested interpretation of the quantum theory in terms of "hidden" variables. I and II. *Physical Review*, 85:166–193, 1952.

Luca Bombelli, Joohan Lee, David Meyer, and Rafael D. Sorkin. Space-time as a causal set. *Physical Review Letters*, 59:521–524, 1987.

Eric A. Brewer. Towards robust distributed systems. In *Proceedings of the 19th ACM Symposium on Principles of Distributed Computing (PODC)*, keynote, 2000.

Daniel M. Chapiro. *Globally-Asynchronous Locally-Synchronous Systems.* PhD thesis, Stanford University, 1984.

Bryce S. DeWitt. Quantum theory of gravity. I. The canonical theory. *Physical Review*, 160:1113–1148, 1967.

Hartmut Ehrig, Karsten Ehrig, Ulrike Prange, and Gabriele Taentzer. *Fundamentals of Algebraic Graph Transformation.* Springer, Berlin, 2006.

Hugh Everett III. "Relative state" formulation of quantum mechanics. *Reviews of Modern Physics*, 29:454–462, 1957.

E. T. Jaynes. Information theory and statistical mechanics. *Physical Review*, 106:620–630, 1957.

Simon Kochen and Ernst P. Specker. The problem of hidden variables in quantum mechanics. *Journal of Mathematics and Mechanics*, 17:59–87, 1967.

Paul Kocher, Jann Horn, Anders Fogh, Daniel Genkin, Daniel Gruss, Werner Haas, Mike Hamburg, Moritz Lipp, Stefan Mangard, Thomas Prescher, Michael Schwarz, and Yuval Yarom. Spectre attacks: Exploiting speculative execution. In *2019 IEEE Symposium on Security and Privacy*, pages 1–19, 2019.

Leslie Lamport. Time, clocks, and the ordering of events in a distributed system. *Communications of the ACM*, 21(7):558–565, 1978.

Rolf Landauer. Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development*, 5(3):183–191, 1961.

Pattie Maes. Concepts and experiments in computational reflection. In *Proceedings of OOPSLA '87*, 1987.

Dag Arne Osvik, Adi Shamir, and Eran Tromer. Cache attacks and countermeasures: The case of AES. In *Topics in Cryptology — CT-RSA 2006*, pages 1–20. Springer, 2006.

Don N. Page and William K. Wootters. Evolution without evolution: Dynamics described by stationary observables. *Physical Review D*, 27:2885–2892, 1983.

Jorma Rissanen. Modeling by shortest data description. *Automatica*, 14:465–471, 1978.

Carlo Rovelli. Relational quantum mechanics. *International Journal of Theoretical Physics*, 35:1637–1678, 1996.

Carlo Rovelli. Forget time. *Foundations of Physics*, 41:1475–1490, 2011.

Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27:379–423, 623–656, 1948.

Brian Cantwell Smith. *Reflection and Semantics in a Procedural Language*. MIT Laboratory for Computer Science, 1982.

John von Neumann. *Mathematical Foundations of Quantum Mechanics*. Princeton University Press, Princeton, 1955. English translation of the 1932 German edition.

Norbert Wiener. *Cybernetics: Or Control and Communication in the Animal and the Machine*. MIT Press, Cambridge, MA, 1948.

Stephen Wolfram. *A New Kind of Science*. Wolfram Media, Champaign, IL, 2002.

Stephen Wolfram. *A Project to Find the Fundamental Theory of Physics*. Wolfram Media, Champaign, IL, 2020.